



C#

Level 1

Array Operation Count,
Sum, Average



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Definition

LINQ Array Operations

 `System.Linq` provides methods for performing operations on arrays and collections.

 `Count()`, `Sum()`, and `Average()` are useful LINQ methods.

Explanation

Count()

 Returns the number of elements in the array.

Sum()

 Returns the total of all elements.

Average()

 Returns the average value of the elements.

 $\text{Average} = \text{Sum} \div \text{Count}$.

💡 Example

```
int[] numbers = { 20, 22, 19, 18, 1 };  
Console.Write("Array elements: {");  
foreach (int number in numbers)  
{  
    Console.Write(number + ", ");  
}  
Console.WriteLine("}");  
Console.WriteLine("Count : " + numbers.Count());  
Console.WriteLine("Sum : " + numbers.Sum());  
Console.WriteLine("Average: " + numbers.Average());
```

➡ Result

Array elements: { 20, 22, 19, 18, 1, }

Count: 5

Sum: 80

Average: 16

Note

- ◆ These methods are LINQ extension methods.
- ◆ Add using `System.Linq;` to use them.
- ◆ `Count()` counts elements, while `Sum()` adds their values.

Comparison

 Count()

✓ Number of elements.

 Sum()

✓ Total of all elements.

 Average()

✓ Average value = $\text{Sum} \div \text{Count}$.

✦✦ Summary

- ✓ Count() → number of elements.
- ✓ Sum() → total of elements.
- ✓ Average() → average value.
- ✓ using System.Linq; is required for these LINQ methods.

Thank you 😊